Output:

--- STEP 1: Data Preparation and Exploration ---

1.1. Downloading Data...

/tmp/ipython-input-913944317.py:40: FutureWarning: YF.download() has changed argument auto_adjust default to True

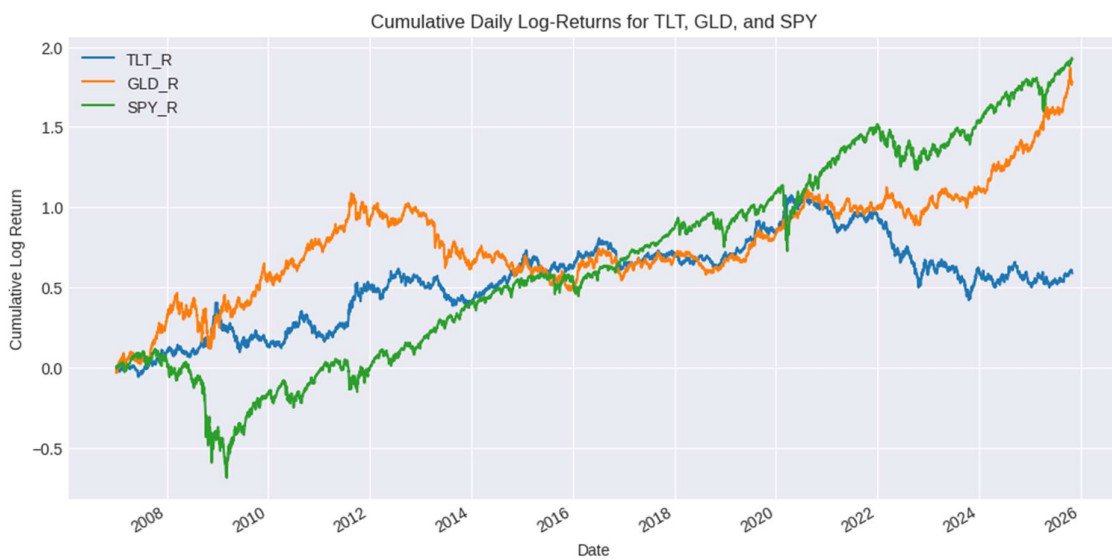
```
data = yf.download(TICKERS, start=START_DATE, end=END_DATE)
```

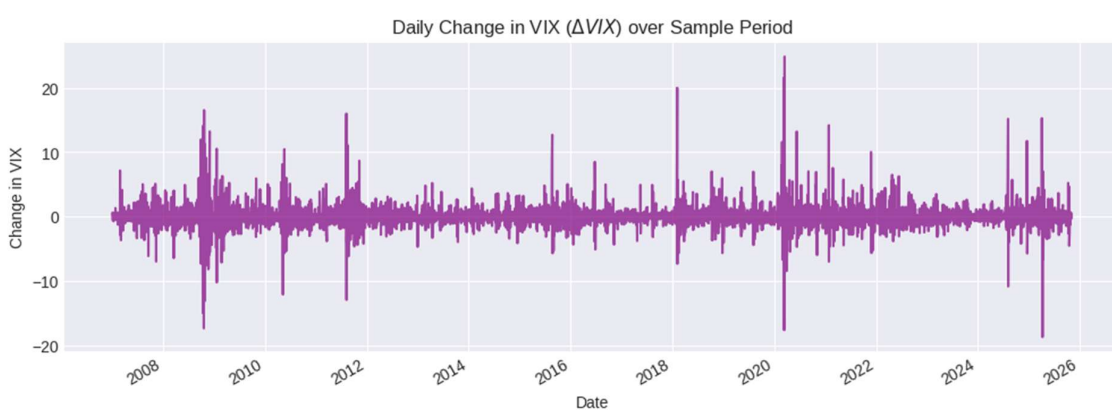
[*********************100%***********************] 4 of 4 completed

1.2. Computing Returns...

1.3. Aligning and Cleaning Data...

Sample Period: 2007-01-04 to 2025-11-03





--- STEP 2 & 3: Modeling, Selection, and Interpretation ---

Fitting 2-State HMM...

Fitting 3-State HMM...

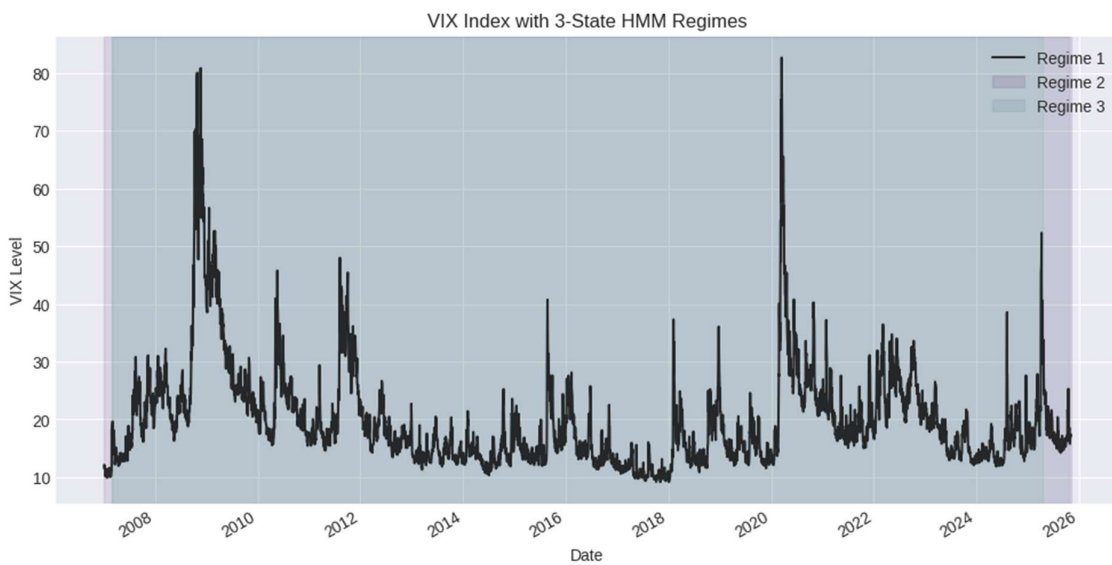
3.1. HMM Comparison Metrics:

	Log-Likelihood	N_Parameters	AIC	BIC
2	-8353.2886	7	16720.5772	16765.8223
3	-8066.7911	14	16161.5822	16252.0723

3.2. Selected Model: 3-State HMM

Transition Probability Matrix (P):

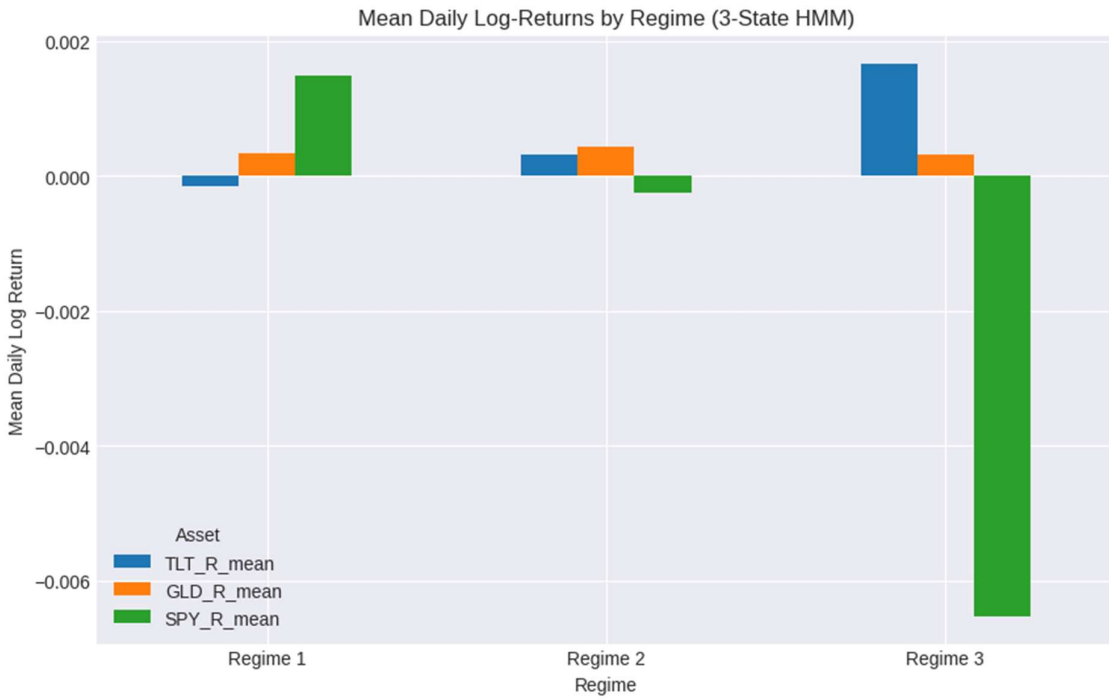
	To R1	To R2	To R3
From R1	0.8741	0.1259	0.0000
From R2	0.0153	0.9039	0.0808
From R3	0.0019	0.0604	0.9377



3.3. State-Conditional ETF Return Statistics:

	TLT_R_mean	TLT_R_std	GLD_R_mean	GLD_R_std	SPY_R_mean	SPY_R_std

Regime 1	-0.0002	0.0080	0.0003	0.0092	0.0015	0.0062
Regime 2	0.0003	0.0100	0.0004	0.0116	-0.0002	0.0130
Regime 3	0.0017	0.0187	0.0003	0.0211	-0.0065	0.0362



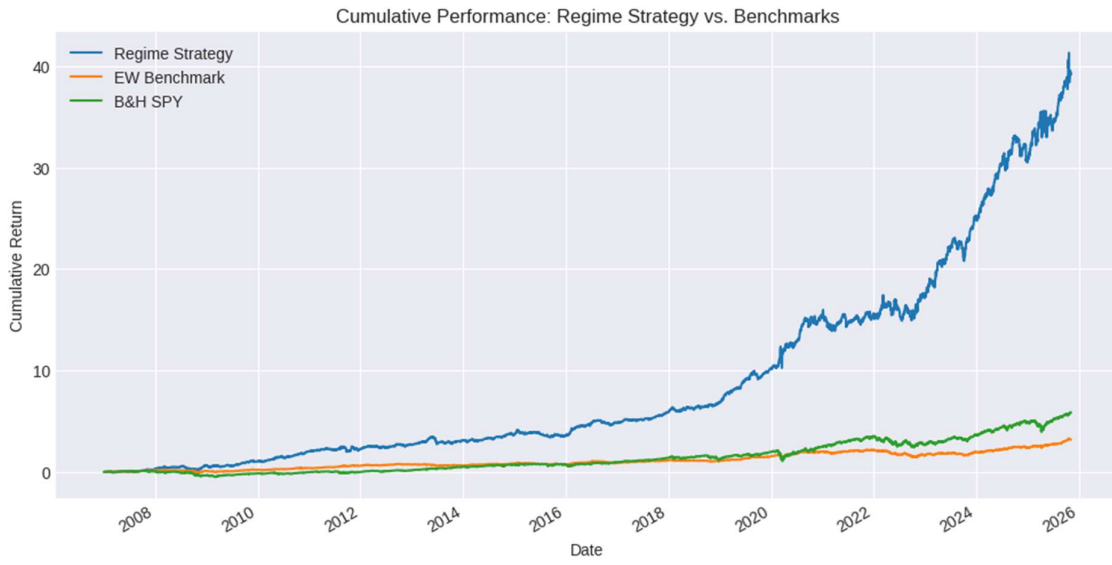
--- STEP 4: Designing the Rotation Strategy ---

4.2. State -> Allocation Mapping (1-Day Lag):

Regime	Chosen Asset (100% Allocation)	Highest Mean (mu)
Regime 1	SPY_R	0.0015
Regime 2	GLD_R	0.0004
Regime 3	TLT_R	0.0017

--- STEP 5: Backtesting and Evaluation ---

5.4. Cumulative Performance Plot:



5.5. Discussion on Sensitivity is required based on the final metrics.